VF-B05 Die C Tear

Test Method for Die C Tear	Test method #: VF- B05 REF #: TM-R-040
Date Issued: November 18, 2022 Revision #: 4	Reason for Revision: 1. Update references 2. Rectify thickness gauge capable of measuring to 0.01mm

PURPOSE AND SCOPE: To measure the load required to tear a material perpendicular to its thickness at the apex of a 90° cut in the test sample.

REFERENCES: ASTM D624

EQUIPMENT AND MATERIAL:

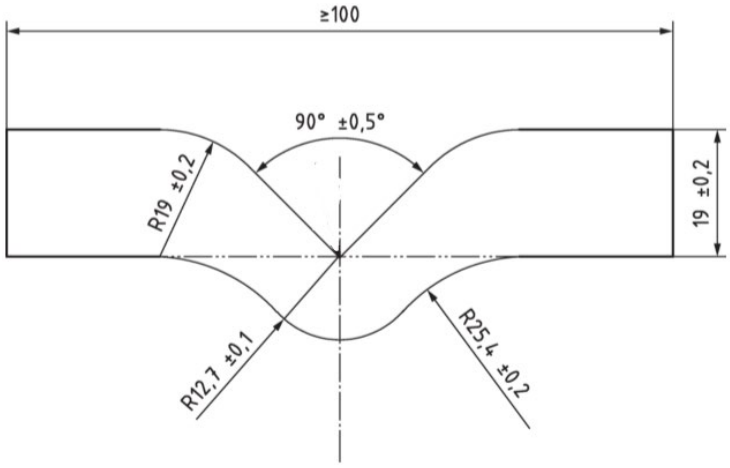
1. Tensile testing machine, meeting the requirements of ASTM D638
2. Clamping jaws, preferably rubber-faced
3. Thickness gauge, capable of measuring to 0.01mm
4. Vernier caliper, capable of measuring to 0.01mm
5. Rubber, plastic, foam

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at standard laboratory environment at 23.0±2.0 ˚C Temperature and 55 ±10 % Relative Humidity.

2. Cut samples using a die as described in **Figure A.** Test samples must be of a uniform thickness, and within the thickness range of 3.0±1.0mm. To ensure the accuracy of this test method, make certain that the cutting edge of the die is sharp and undamaged.

Figure A



VF-B05 Die C Tear

REPETITIONS: Test three (3) samples per material.

TEST PROCEDURE:

Test should be carried out at standard laboratory environment @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Measure the thickness of the test sample at the apex of the 90°cut, and record to the nearest 0.1mm (**T**).
2. Zero and calibrate the testing machine.
3. Set the tensile testing machine to a crosshead speed of 500 mm/minute and a jaw gap of 57+/-3mm.
4. Insert the sample centrally in the top and bottom jaws of the tensile testing machine such that the sample is vertical, no slack or tension is present, and equal portions of the test sample are positioned in the top and bottom jaws.
5. Operate the tensile testing machine at the specified crosshead speed of 500 mm/minute.
6. Record the tearing strength as the load at the instant of the initial tear to the nearest 0.1kgf (L). ***The initial tear is generally the highest peak obtained because this is where the failure occurs. Stop the test once the tearing already established.***

CALCULATION:

Die-C tear strength (kgf/cm) = L/T

Note: For a few materials which are typically supplied in a specified thickness (e.g. “thermo” heel counters and toe box), the thickness of the test material may not be factored into the calculation. With these materials, the die-C tear strength is simply reported as the tearing strength (**L**) in kgf.

REPORT:

The average of the three die-C tear strength measurements to the nearest 0.1kgf/cm

For thermo heel counter and toe box, report in kgf